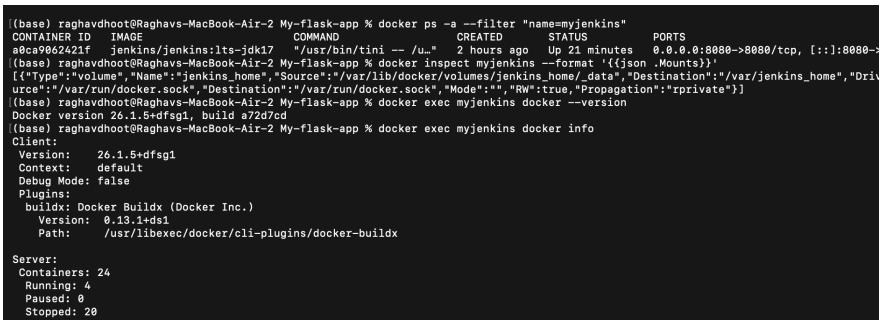
SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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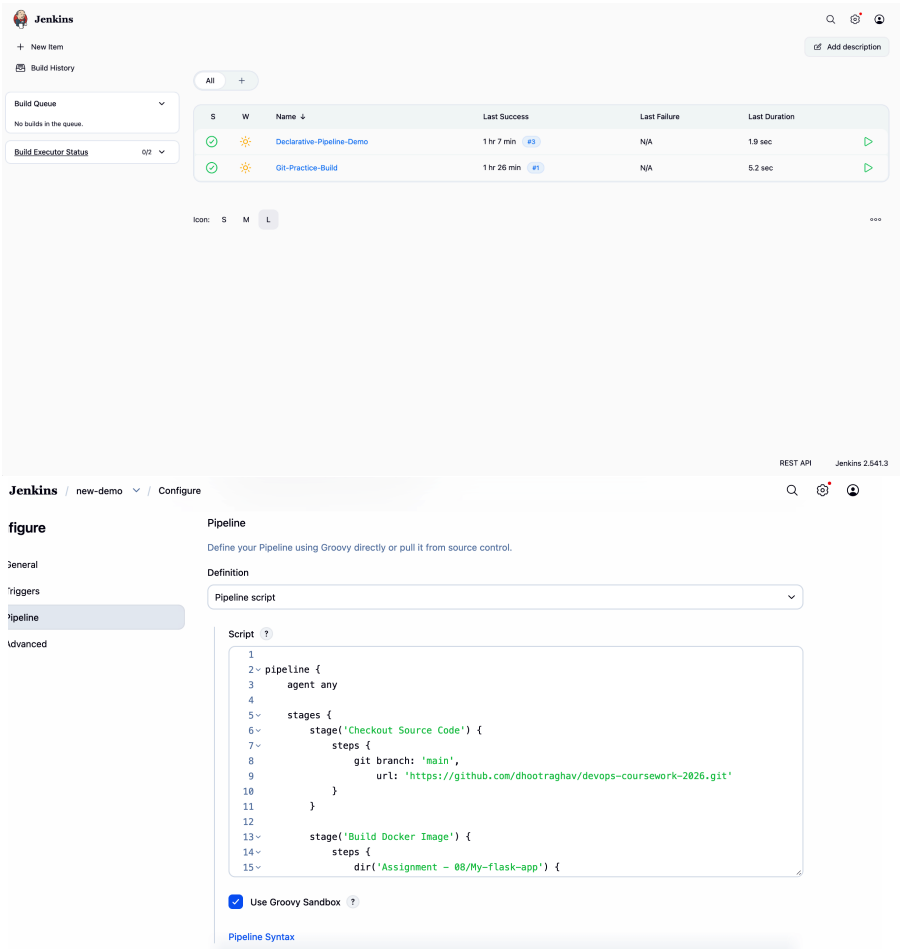
Department of Computer Science and Engineering
DevOps Lab - L3

Lab Assignment No: 14	
Name of Student	Raghav Dhoot
PRN	23070122170
Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	CI/CD — Jenkins Pipeline building Docker Image
Implementation & Theory	 Jenkins container recreated with Docker access — Stopped and removed the existing Jenkins container, then re-ran it with the host's Docker socket mounted (-v /var/run/docker.sock:/var/run/docker.sock), since Jenkins runs inside its own container and has no Docker daemon of its own.

```
(base) raghavdhoot@Raghavs-MacBook-Air-2 My-flask-app % docker exec -u root myjenkins sh -c "apt-get update && apt-get install -y docker.io"
Hit:1 http://deb.debian.org/debian trixie InRelease
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian-security trixie-security/main arm64 Packages [267 kB]
Fetched 357 kB in 0s (1260 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
docker.io is already the newest version (26.1.5+dfsg1-9+deb13u1).
0 upgraded, 0 newly installed, 0 to remove and 49 not upgraded.
(base) raghavdhoot@Raghavs-MacBook-Air-2 My-flask-app % docker exec myjenkins docker --version
Docker version 26.1.5+dfsg1, build a72d7cd
(base) raghavdhoot@Raghavs-MacBook-Air-2 My-flask-app %
```

Docker CLI installed inside Jenkins — Installed docker.io inside the running Jenkins container via apt-get, then verified with docker --version that the CLI could talk out through the mounted socket.

Repo pushed — Committed and pushed local work (assignments 8-13) to the Devops-Coursework GitHub repo so Jenkins had something to check out and build.



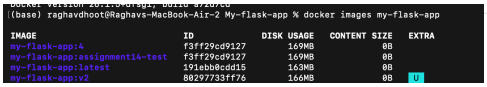
Pipeline updated with Build Docker Image stage — Added a new stage to the existing Declarative-Pipeline-Demo pipeline that

changes into the Assignment - 8/my-flask-app directory and runs docker build -t my-flask-app:${BUILD_NUMBER} ., auto-tagging the image per build.

Console Download Copy View as plain text

```
Started by user Raghav Dhoot
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/jenkins_home/workspace/new-demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout Source Code)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > git rev-parse --resolve-git-dir /var/jenkins_home/workspace/new-demo/.git # timeout=10
Fetching changes from the remote Git repository
 > git config remote.origin.url https://github.com/dhootraghav/devops-coursework-2026.git # timeout=10
Fetching upstream changes from https://github.com/dhootraghav/devops-coursework-2026.git
 > git --version # timeout=10
 > git --version # 'git version 2.47.3'
 > git fetch --tags --force --progress -- https://github.com/dhootraghav/devops-coursework-2026.git +refs/heads/
*:refs/remotes/origin/* # timeout=10
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 1beafa4866b487f2c7dc12b3e859b0f774a6bf4a (refs/remotes/origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f 1beafa4866b487f2c7dc12b3e859b0f774a6bf4a # timeout=10
 > git branch -a -v --no-abbrev # timeout=10
 > git branch -D main # timeout=10
 > git checkout -b main 1beafa4866b487f2c7dc12b3e859b0f774a6bf4a # timeout=10
Commit message: "Add files via upload"
 > git rev-list --no-walk 1beafa4866b487f2c7dc12b3e859b0f774a6bf4a # timeout=10
[Pipeline] }
```

```
REPOSITORY     TAG                IMAGE ID       CREATED         SIZE
my-flask-app   4                  f3ff29cd9127   7 minutes ago   169MB
my-flask-app   assignment14-test  f3ff29cd9127   7 minutes ago   169MB
my-flask-app   v2                 80297733ff76   4 hours ago     166MB
my-flask-app   latest             191ebb0cdd15   5 weeks ago     163MB
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Assignment 14 Pipeline SUCCESS!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



Successful build — Re-ran the pipeline; all stages completed, and the Build Docker Image stage successfully built and tagged my-flask-app:5, with the full pipeline finishing SUCCESS.